Sketch2Face

Cross-modal face retrieval
CUFS CUHK subset

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Guide: [Guide name - placeholder]

Problem and objective

Input: one face sketch and a gallery of candidate photos

Output: photos ranked by similarity to the person in the sketch

Challenge: preserve identity across pencil strokes and photographic appearance

Objective: build the matcher and measure rank-k retrieval accuracy

Dataset audit

Full CUFS description: 606 identities across CUHK, AR, and XM2VTS

Downloaded mirror: 188 usable CUHK sketch-photo pairs

Extra unpaired sketches and duplicate original files were excluded

Every reported result is for the verified CUHK subset

Identity-disjoint evaluation

114 training identities: learn sketch-photo correspondence

37 validation identities: select the checkpoint

37 test identities: final held-out experiment

Each test sketch searches all 37 held-out photos

Model architecture

Stage	Operation	Output
Preprocessing	Grayscale, resize, normalize	3 x 224 x 224
Modality input	Sketch or photo residual adapter	3 x 224 x 224
Shared encoder	ImageNet-pretrained ResNet-18	512 features
Projection	Linear, ReLU, Linear, normalize	256-D embedding
Retrieval	Cosine similarity to gallery	Ranked photos

Paired contrastive learning

A batch contains sketches and photos of distinct paired identities

Correct sketch-photo pairs are positives; other people are negatives

Learn in both directions: sketch to photo and photo to sketch

$L = 0.5 \times [CE(S\ P^T / \tau) + CE(P\ S^T / \tau)],\ \ \tau = 0.07$

Training and comparisons

20 epochs; batch size 32; AdamW optimization

Freeze backbone for 3 epochs, then fine-tune the final residual stage

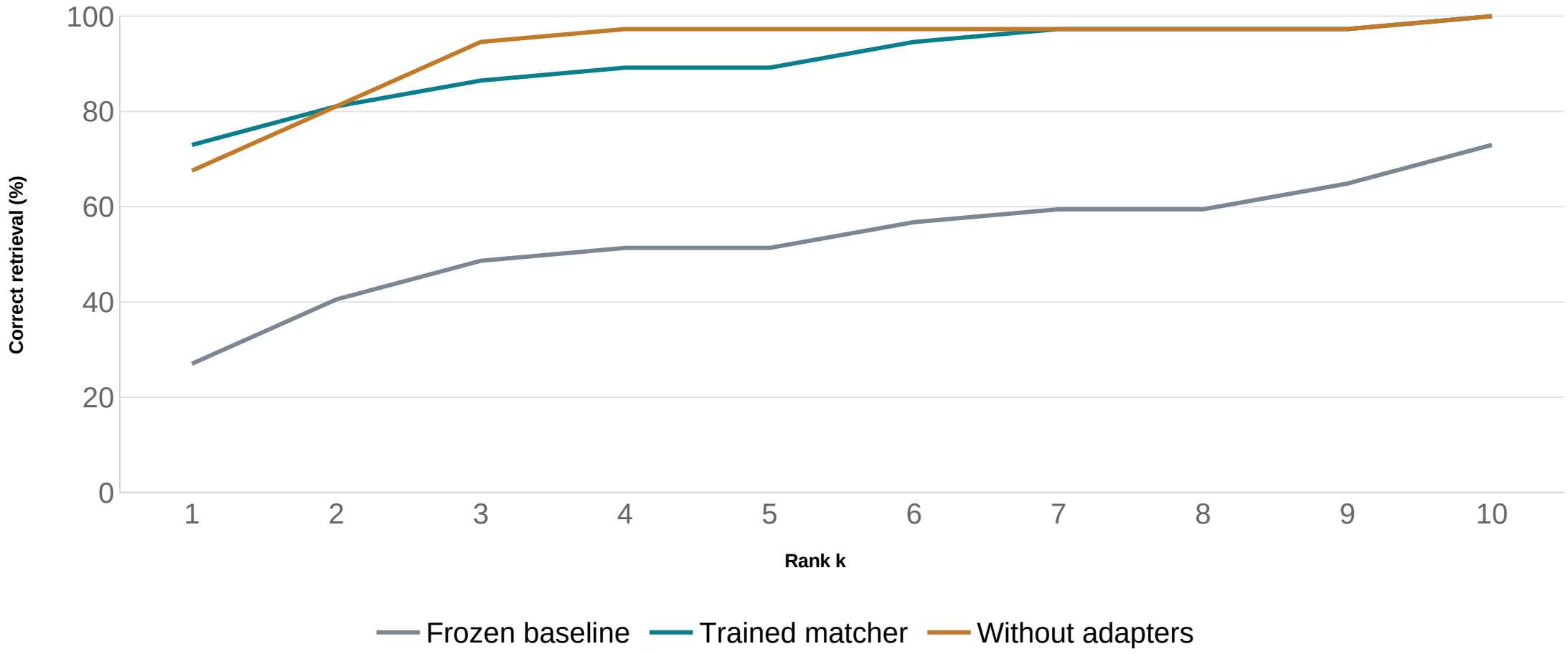
Compare frozen baseline, trained matcher, and no-adapter ablation

Repeat matcher training with seeds 42, 43, and 44

Measured retrieval accuracy

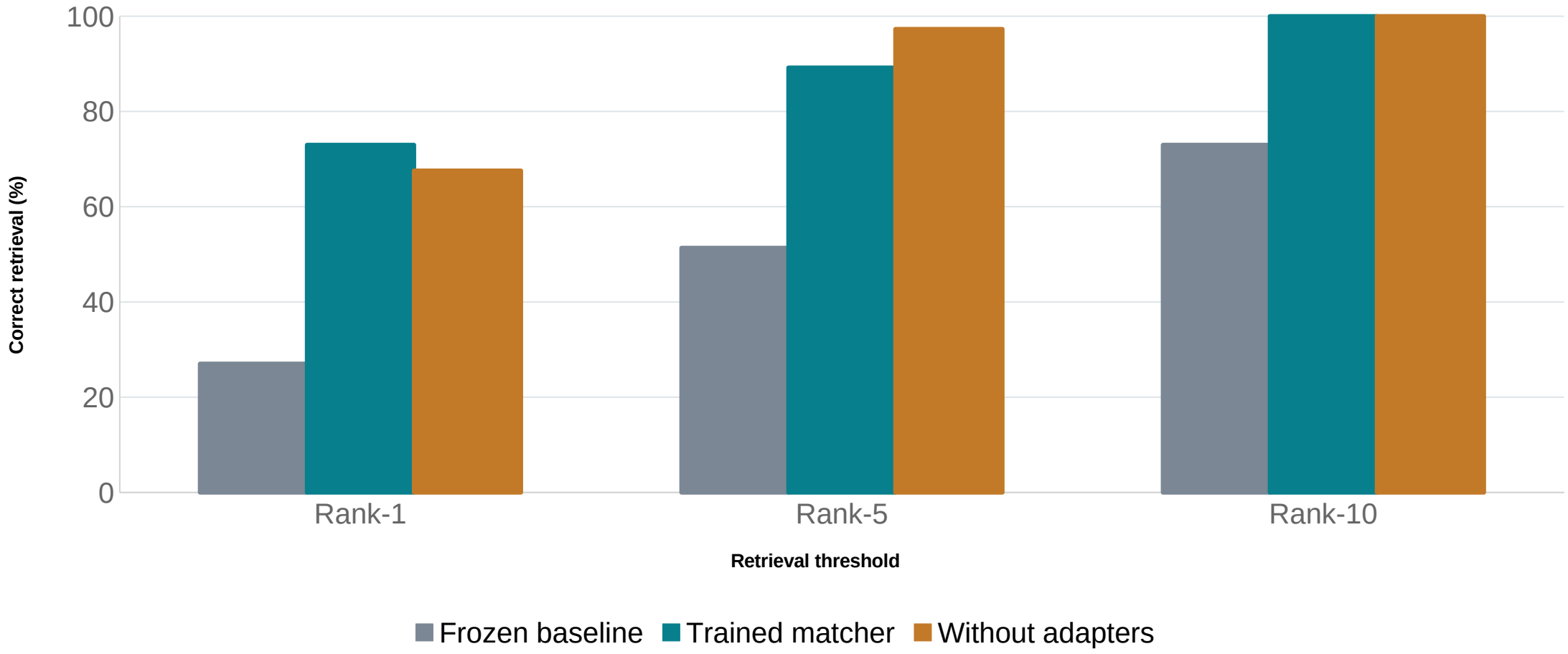
Model	Epoch	Rank-1 (%)	Rank-5 (%)	Rank-10 (%)
Frozen baseline	0	27.03	51.35	72.97
Matcher seed 42	20	72.97	89.19	100.00
Matcher seed 43	20	67.57	89.19	100.00
Matcher seed 44	16	67.57	91.89	100.00
Without adapters	18	67.57	97.30	100.00

CMC comparison



37 test sketches searching the same 37-photo gallery

Rank-k comparison



37 test sketches searching the same 37-photo gallery

What the experiments show

Primary rank-1: 27 correct queries out of 37 (72.97%)

Frozen baseline: 10 correct queries out of 37 (27.03%)

Paired rank-1 gain: 45.95 percentage points; 95% interval 27.03–64.86

Three-seed mean rank-1: 69.37% ± 3.12 percentage points

Adapter ablation and errors

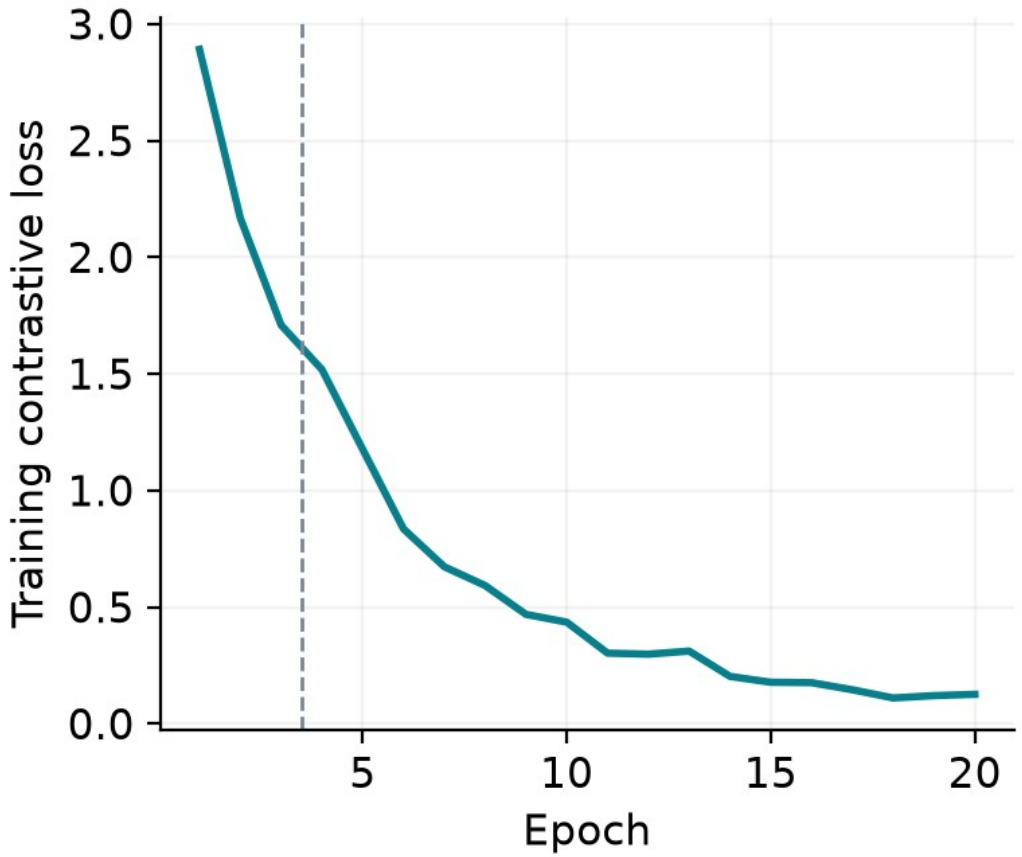
Without adapters: 67.57% rank-1, 97.30% rank-5

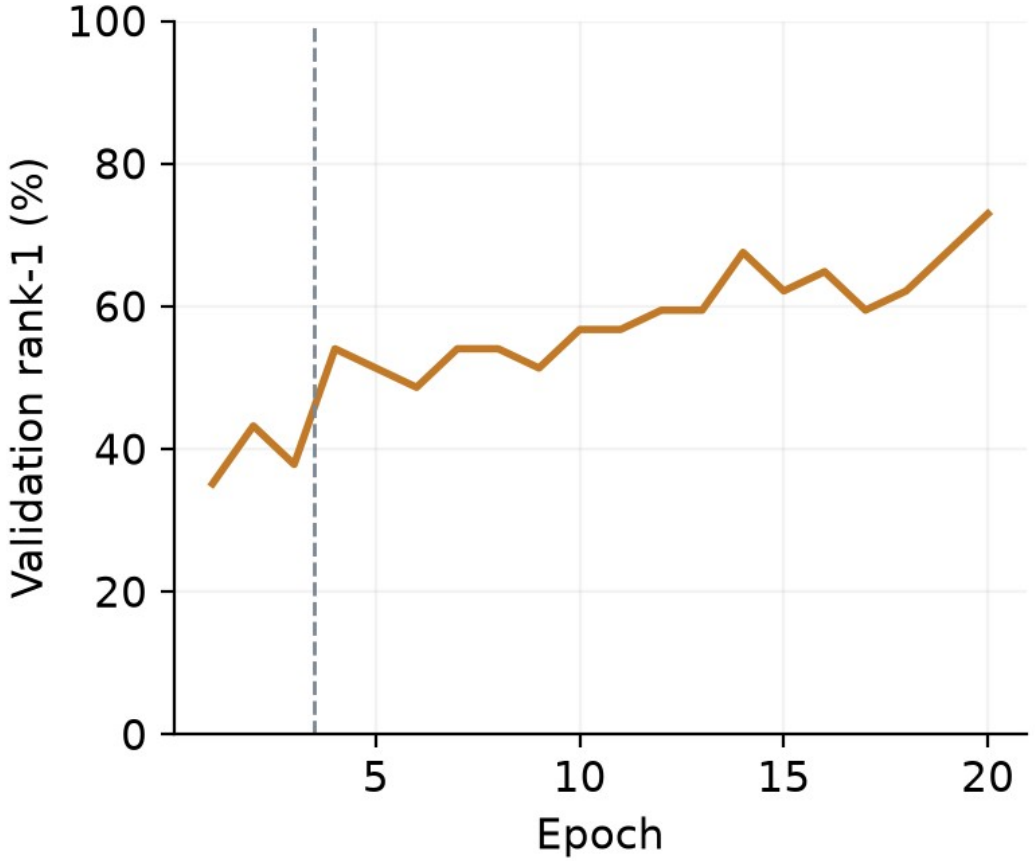
Adapters improve primary rank-1, but reduce rank-5 in this comparison

Ten primary queries fail at rank 1; the hardest correct match is rank 10

One ablation seed does not establish a consistent adapter benefit

Learning curves





Limitations

Only the CUHK subset; AR and XM2VTS pairs remain untested

Small closed-set gallery: 37 photos; no unknown-person rejection

Viewed artist sketches differ from eyewitness and composite sketches

No demographic fairness, forensic reliability, or state-of-the-art claim

Demo and conclusion

Website: http://localhost:8000

Upload a sketch and candidate photos, then inspect the ranked results

Paired training improves the measured frozen baseline

Next: full CUFS, larger galleries, repeated ablations, forensic datasets

References

Wang & Tang, Photo-Sketch Synthesis and Recognition, TPAMI 2009

Zhang et al., Coupled Information-Theoretic Encoding, CVPR 2011

He et al., Deep Residual Learning, CVPR 2016

Radford et al., Contrastive multimodal learning, ICML 2021

Loshchilov & Hutter, AdamW, ICLR 2019; CUFS mirror, Kaggle v1